

Emmanouil “Manos” Farsarakis

Systems Architect & AI Researcher — Intel

Edinburgh, United Kingdom · Eligible to work in the US, UK & EU

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PROFILE

Systems architect specializing in AI accelerator and memory-system architecture evaluation: translating hardware specifications into expected real-world workload performance, and validating those projections with hands-on benchmarking and profiling. A decade at the intersection of AI workloads and hardware design — early accelerator characterization (Intel PiUMA), dataflow-vs-GPU architecture comparison for inference serving, low-precision (int8) inference, and system-level performance projection as technical lead for SPEC Cloud simulation. Track record of bridging internal architecture teams and external partners (Harvard, University of Edinburgh, technology vendors), with peer-reviewed systems research (IEEE ISPASS) and a US patent. The through-line: quantifying the FLOP/s-versus-TB/s balance at every level of the memory hierarchy, and turning measurement into evidence that steers silicon decisions.

EXPERIENCE

Intel

Nov 2023 – Present

AI Systems & Solutions Engineer, Systems Architecture & Engineering — Edinburgh (remote)

- Characterized VLA serving ($\pi 0.5$) across model structure, attention behavior, and BF16/INT8 execution; mapped compute, memory, latency, and power demands onto candidate architectures, comparing dataflow accelerators with heterogeneous GPUs and fixed-function inference ASICs on power efficiency (FLOPS/W), latency, and throughput.
- Technical lead for SPEC Cloud benchmark simulation and system-level performance projection, leading a team of five — aligning measured behavior against modeled projections to validate architecture decisions before commitment.
- Drive inference-focused architecture research for large-memory-capacity serving: evaluated compute throughput, memory bandwidth constraints, and interconnect scaling across candidate technologies including silicon photonics and rack-scale optical interconnects, disaggregated and in-network memory, and KV-cache placement across heterogeneous memory tiers.
- Assessed emerging vendor technologies (e.g. Celestial AI optical interconnect) for memory-centric inference, translating vendor specifications into projected workload-level impact across on-chip vs off-chip solutions.

Intel

Feb 2022 – Nov 2023

Artificial Intelligence Researcher — Edinburgh (remote)

- Coordinated research with leading scientists on next-generation AI algorithms — GNNs, NLP, and recommendation systems — analyzing how workload behavior (sparsity, memory access patterns, data movement) interacts with accelerator design choices.
- Established and led the Harvard × Intel collaboration on graph-learning acceleration (~7 contributors across Intel, Intel Labs, and Harvard SEAS): owned inception, stakeholder management, software development, and performance optimization — resulting in an IEEE ISPASS 2023 paper, a journal paper (TASK Quarterly 2024), and a US patent.

Intel

Feb 2019 – Feb 2022

Platform Architect — Edinburgh (hybrid)

- Led AI workload characterization for the Intel PiUMA sparse accelerator during early evaluation: profiling, microbenchmarking, and scalability analysis of graph neural network workloads against the architecture's compute and memory-system characteristics (published at IEEE ISPASS 2023).
- Drove novel AI optimizations — SIMT execution, sparse-compute acceleration, and quantization — quantifying precision/performance trade-offs for AI workloads.
- Embedded at the University of Edinburgh collaborating with K. Heafield and P. Boyle: co-developed CPU-efficient, low-precision (int8) Marian neural machine translation, winner of the 2020 WNGT efficiency task; the underlying int8 GEMM kernels were adopted into SPEC CPU (2026), where Marian is the sole NLP workload.

The University of Edinburgh — EPCC

2014 – 2019

Applications Consultant / Developer — Edinburgh

- HPC performance engineering across finance, cyber security, and exascale projects: porting, profiling, and optimization on academia–industry collaborations.

- Built system-level workload characterization of ARCHER (UK national supercomputer) and researched data-aware HPC scheduling for Intel 3D XPoint non-volatile memory (NextGenIO).

SELECTED PROJECTS

Inference Architectures for VLA Robotics — Extended an AI roofline simulator to cover new accelerator architectures and model types; characterized VLA serving ($\pi 0.5$) and compared dataflow accelerators, heterogeneous GPUs, and fixed-function inference ASICs on FLOPS/W, latency, and throughput.

Silicon Photonics & Disaggregated Memory — Evaluated rack-scale optical interconnects and disaggregated/in-network memory for inference serving; compared latency, throughput, and cost of on- vs off-chip memory solutions from vendor specifications.

Sparse Computational Infrastructure for GNNs (Harvard $\times$ Intel) — Cross-organization accelerator-evaluation collaboration; ISPASS 2023, journal paper, US patent.

Low-Precision CPU Inference — Marian NMT in SPEC CPU — int8 inference kernels; 2020 WNGT efficiency-task winner; adopted into SPEC CPU 2026.

SELECTED PUBLICATIONS & PATENTS

- Characterizing the Scalability of Graph Convolutional Networks on Intel® PIUMA — IEEE ISPASS 2023.
- Neural Network training and inference with Hierarchical Adjacency Matrix — US Patent Application 18/325,348 (2023).
- Optimizing Graph Learning using Hierarchical Graph Adjacency Matrix (HGAM) — TASK Quarterly 28(3), 2024.
- Edinburgh's submissions to the 2020 machine translation efficiency task — WNGT 2020.
- Full list: Google Scholar.

EDUCATION

MSc in High Performance Computing

2014

The University of Edinburgh — Distinction; Class Representative. ISC'14 Student Cluster Competition (as MSc student, June 2014): designed and optimized the software stack of a custom GPU cluster — Highest Linpack award (3.38 Tflops/kW, est. 4th on the June 2014 Green500); ported GADGET-3 kernels to GPU via OpenACC for $2\times$ speedup.

BSc Physics (Computational Science specialization)

2011

University of Crete — Top 5% of class